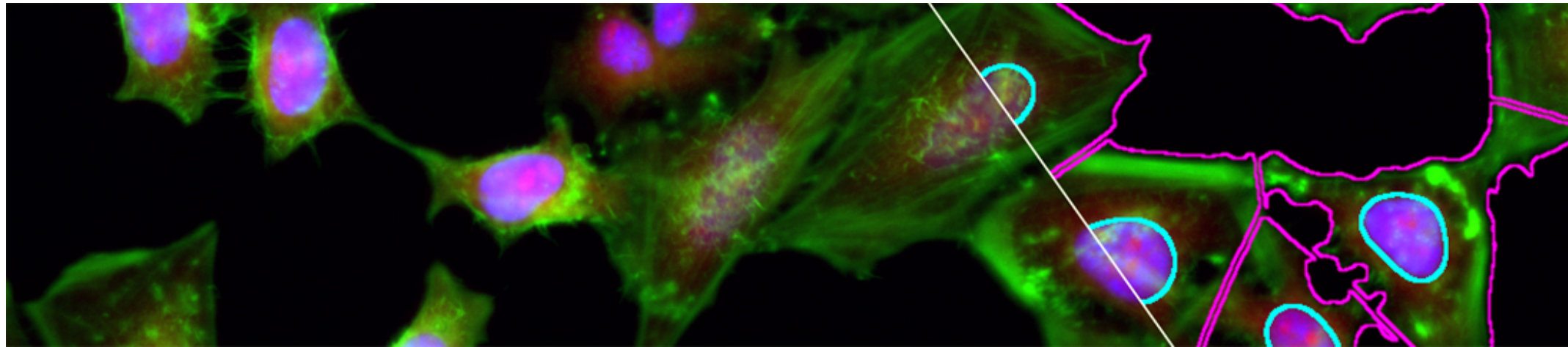


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2.1.1. Description & Outline



1. Digital Images

2. Of Colors & Dimensions

3. Resizing, Interpolation & Mathematical Operations

4. Filtering

5. Segmentation, from Intensities to Objects

6. Regions of Interest & Results

7. Color in Depth, Projections & Reslicing

Extra. ImageJ Macro Language Programming Primer

Week Overview

- Section 1: Introduction
- Section 2: Seeing Colours
- Section 3: Image Bit-Depth & Channels
- Section 4: Beyond Channels
- Section 5: Conclusion

Description

This week, you will see how color can be assigned to pixel intensities in order to enhance visual contrast. This is the concept of lookup tables. You will learn what image bit-depth represents about special images types, like RGB images and how to work with multichannel data. Finally we go beyond single channel images and cover the many dimensions that scientific image data can have. This will be followed by a short discussion on what the limits are for microscopy acquisitions, where sample health, data size, image quality, and number of image dimensions lead to the need for compromises.

By the end of this week, you will be able to use lookup tables in Fiji and will be able to cite a few commonly used ones. You'll know how to recognize and explain the use for different bit-depths. With the arrival of multi-dimensional data, you will be able to distinguish the 5 (five) most common dimension of an image and how to navigate them in Fiji. Finally you'll be able to reflect on the technical considerations when designing a microscopy experiment which might involve a lot of images and dimensions.

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